Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/24/2006
PI Name:	Pavalko , Fredrick M Ph.D.		
Project Title:	Regulation Of Osteoblast Apoptosis By Fluid Shear Stress		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	46202-5120	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	12/01/2002	**End Date:**	11/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1606		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Bone mass is increased in response to mechanical loading, while chronic unloading of bone, such as occurs during prolonged bed rest and in microgravity during space flight, results in atrophy of bone. Exposure of osteoblasts and osteocytes to fluid shear stress (FSS), induced by the movement of interstitial fluid within the spaces inside bone, has been hypothesized to play a role in the response of bone to mechanical loading. This hypothesis is supported by in vitro studies demonstrating that bone-forming osteoblasts and osteocytes are stimulated by FSS. Among the responses of bone cells to FSS are the release of autocrine/paracrine signaling factors including prostaglandins and nitric oxide (NO), both of which are known to play a role in bone growth and remodeling. In this project we are seeking to gain a better understanding of the cellular and molecular events that mediate the response of bone to loading. We are examining the effects of steady and pulsatile FSS on growth and survival of osteoblasts. Our results to date suggest that FSS promotes osteoblast survival in vitro. These results raise the possibility that mechanical loading may promote bone formation by increasing the lifespan of osteoblasts in bone through inhibition of osteoblast apoptosis. Our primary objective is to test		

	the hypothesis that FSS plays a role in regulating osteoblast survival using an in vitro cell culture model. Identify cellular mechanisms that regulate apoptosis in bone cells. Inhibition of osteoblast apoptosis could improve bone growth and remodelling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the past year we have continued to focus on trying to understand the cellular mechanisms that mediate the anti-apoptotic effects of fluid shear stress. We have continued to focus our efforts on understanding the effects of steady and pulsatile fluid shear stress on osteoblasts. We recently submitted a manuscript comparing directly the effects of steady unidirectional shear versus osciallatory shear (Ponik, S.M., Triplett, J.W. and Pavalko, F.M. Osteoblasts and Osteocytes Respond Differently to Oscillatory and Unidirectional Fluid Flow Profiles. Manuscript submitted to Bone, December 2005). We have also published a study describing the beneficial effects of etoposide on adenoviral-mediated protein expression in osteoblasts, a method that has improved our ability to express exogneous proteins in osteoblasts, a procedure that has been notoriously difficult (Triplett, J.W., Herring, B.P. and Pavalko, F.M. (2005) Adenoviral transgene expression enhanced by co-treatment with etoposide in cultured cells. Biotechniques, 39:826-832). Recent studies by Sakata et al. (2004, JBMR, 19: 436-446) found that skeletal unloading of mice induced IGF-1 “resistance” in part by downregulation of integrin signaling. Similarly, we found that insulin-like growth factor-1 (IGF-1) is significantly more effective at preventing TNF-a-induced apoptosis when cells are first subjected to mechanical loading by exposure to fluid shear stress than if cells are maintained in culture under static conditions before receiving IGF-1. We have continued to focus on understanding the cellular mechanisms responsible for this “sensitization” of IGF-1 receptors by fluid shear stress. To that end, we have recently submitted a manuscript describing the effect of fluid shear stress on signaling via the IGF-1 receptor and the beneficial effects of shear on IGF-1-mediated inhibition of apoptosis in osteoblasts (Pavalko, F.M., Gerard-O’Riley, R., Tekulve, K., and Norvell, S.M. Fluid Shear Stress Enhances IGF-1 Receptor Signaling in Osteoblasts as a Mechanism to Inhibit Apoptosis. Manuscript submitted). Interestingly, sensitization of the IGF-1 receptor by fluid shear stress is blocked when PKC-zeta activity is inhibited in osteoblasts. This result suggests that regulation of PKC-zeta activity by fluid shear stress may play a role in regulating IGF-1-mediated pro-life signals in bone cells. Together, our results suggest that fluid shear stress may, in part, regulate apoptosis in osteoblasts via a PKC-zeta-dependent sensitization of IGF-1 receptor signaling. We have also been investigating the effects of altered cell-substrate attachments on basal levels of apoptosis in osteoblasts, as well as on the susceptibility of bone cells to apoptotic stimuli. This work has led to one published manuscript (Ponik, S.M. and Pavalko, F.M. (2004) Formation of Focal Adhesions on Fibronectin Promotes Fluid Shear Stress Induction of COX-2 and PGE2 Release in MC3T3-E1 Osteoblasts. J. Appl. Physiol., 97:135-142.) and a second manuscript that is in preparation and should be submitted during 2006 (Triplett and Pavalko, manuscript in preparation).
Bibliography Type:	Description: (Last Updated: 06/25/2007)
Articles in Peer-reviewed Journals	Triplett JW, Herring BP, Pavalko FM. "Adenoviral transgene expression enhanced by co-treatment with etoposide in cultured cells." Biotechniques. 2005 Dec;39(6):826, 828, 830, passim. PMID: 16382899 , Dec-2005
Articles in Peer-reviewed Journals	Charoonpatrapong K, Shah R, Robling AG, Alvarez M, Clapp DW, Chen S, Kopp RP, Pavalko FM, Yu J, Bidwell JP. "HMGB1 expression and release by bone cells." J Cell Physiol. 2006 May;207(2):480-90. PMID: 16419037 , May-2006
Articles in Peer-reviewed Journals	Norvell SM, Alvarez M, Bidwell JP, Pavalko FM. "Fluid shear stress induces beta-catenin signaling in osteoblasts." Calcif Tissue Int. 2004 Nov;75(5):396-404. PMID: 15592796 , Nov-2004